

MATH 770 - Baire Category Theorem Exposition

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1 Main Theorem

1.1 Motivation

In the period from 1870 to 1880, Cantor was developing point-set theory. Cantor's main objects of study were point-sets in the " n -dimensional arithmetic continuum", which we now know as Euclidean n -space. In general, Cantor did not make contributions to topology with point-set theory, but the theory eventually came to comprise the basis of point-set topology. Cantor, in his point-set language, presented fundamental concepts such as limit points and presented the first published proof of the Bolzano-Weierstrass Theorem.

By the 1890s, younger French analysts such as Borel and Lebesgue were using Cantor's framework to study measurable sets, integration, and the regularity of "definable" sets of reals. Their work on Borel sets and Lebesgue measure aimed to make precise phrases like "almost everywhere" and to control the size of exceptional sets where a function misbehaves. This environment of Cantorian sets, plus emerging measure theory—is the immediate background for Baire.

In 1899 Rene-Louis Baire published his doctoral dissertation *Sur les fonctions de variables reelles*. The thesis originally focused on the classification of "sizes" of sets on the "continuum" or the real line. In particular, he studied functions of one and several real variables, focusing on continuity, discontinuity, and the representations of limits. In order to classify functions whose "bad behavior" is confined to a small exceptional set, Baire introduces the concept of nowhere dense sets and the distinction between first and second category, a distinction we will go over later, and proves that that $\mathbb{R}^n$ cannot be expressed as the countable union of nowhere dense sets.

In the next decade, Baire's presentation is gradually generalized to the context of measure spaces. The presentation I will give in this paper is the generalized form. Before I present the Baire Category Theorem we must go through some preliminary definitions.

1.2 Preliminaries

Definition 1.1 (Metric Space). *A metric space is a set X associated to a distance function $d : X \times X \rightarrow \mathbb{R}$ satisfying the following axioms. Let $x, y, z \in X$*

- (i) *If $x \neq y$ then $d(x, y) > 0$, $d(x, y) = 0$ if and only if $x = y$.*
- (ii) *The distance from one point to another is the same if you reverse directions, $d(x, y) = d(y, x)$*
- (iii) *$d(x, z) \leq d(x, y) + d(y, z)$*

Definition 1.2. *A sequence x_n of points in metric space (X, d) is Cauchy if for all $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that when $m, n \geq N$ it holds that $d(x_n, x_m) < \epsilon$.*

Definition 1.3. *A metric space (X, d) is considered complete if every Cauchy sequence converges to a point in X .*

Definition 1.4 (Open Ball). *Let (X, d) be a metric space. An open ball centered at $x \in X$ or radius $r \in \mathbb{R}$ is a set*

$$B(x, r) := \{y : d(x, y) < r\}$$

A metric space is a particular case of a topology. As such, we can define metric space specific topological constructions.

Definition 1.5 (Open Set). *A set U is open in a metric space (X, d) if, for all $x \in U$, there exists an $\epsilon > 0$ such that $B(x, \epsilon) \subset U$.*

With this definition, we are immediately equipped to prove the following lemma.

Lemma 1.1. *An open ball in a metric space (X, d) is open.*

Proof. Let $B(x, r)$ be a ball in a metric space (X, d) . For all $y \in B(x, r)$ take $\epsilon = \frac{r-d(x,y)}{2} > 0$. Let $B(y, \epsilon)$ be an open ball in (X, d) then for all $z \in B(y, \epsilon)$

$$d(x, z) \leq d(x, y) + d(y, z) \quad (1)$$

$$< d(x, y) + \frac{r - d(x, y)}{2} \quad (2)$$

$$= \frac{r + d(x, y)}{2} < r \quad (3)$$

Therefore $B(y, \epsilon) \subset B(x, r)$. $\square$

Definition 1.6 (Closed Set). *A set U is closed in a metric space (X, d) if $E^c = X - E$ is open in X .*

With this we can now define some of the most important "pieces" of sets in a metric space. Given a set E in a metric space (X, d) we talk about the interior, loosely the "inside" of a set, and the closure, a set plus whatever construction is necessary to act as a "cap" and close the set. In reality a metric space is a particular example of a topological space and the concepts of interior and closure follow from this more general environment.

Definition 1.7 (Interior). *Given a set A in a metric space (X, d) we can define its interior*

$$A^\circ := \bigcup \{U : U \subset A \text{ and } U \text{ is an open subset of } X\}$$

We can reinterpret this definition using our metric

$$A^\circ := \{x : \exists \epsilon > 0, B(x, \epsilon) \subset A\}$$

Definition 1.8 (Closure). *Given a set A in a metric space (X, d) we can define its closure*

$$\bar{A} := \bigcap \{E : A \subset E \text{ and } E \text{ is a closed subset of } X\}$$

We can reinterpret this definition using our metric

$$\bar{A} := \{x : \forall \epsilon > 0, B(x, \epsilon) \cap A \neq \emptyset\}$$

With this information, we can begin to tackle the concept of density.

Definition 1.9. *A set $E \subseteq X$ is dense in X if for any open non-empty subset $U \subseteq X$ that $U \cap E \neq \emptyset$. Equivalently we can say that E is dense if $\bar{E} = X$.*

Before presenting the categories given by Baire, we first need to define the notion of a nowhere dense set.

Definition 1.10 (Nowhere Dense). *A set E is nowhere dense if $(\bar{E})^\circ = \emptyset$.*

A particularly useful property that arises from this definition is that

Lemma 1.2. *A set is nowhere dense if and only if its complement contains a dense open set.*

Proof. Let A be a nowhere dense set in X . We utilize the following facts: For any $B \subseteq X$, $X \setminus \bar{B} = (X \setminus B)^\circ$, $X \setminus B^\circ = \overline{X \setminus B}$. Then,

$$(\bar{A})^\circ = \emptyset \iff X \setminus (\bar{A})^\circ = X \quad (4)$$

$$\iff \overline{X \setminus \bar{A}} = X \quad (5)$$

$$\iff \overline{(X \setminus A)^\circ} = X \quad (6)$$

$$\iff (X \setminus A)^\circ \text{ is dense in } X \quad (7)$$

Since $(X \setminus A)^\circ \subseteq X \setminus A$ then $X \setminus A$ admits a dense open subset. To reverse the process suppose that $X \subset A$ contains a dense open subset D , then $D \subseteq (X \setminus A)^\circ$ implies that $(X \setminus A)^\circ$ is dense and we can move backwards through our if and only if statements to get the desired result. $\square$

With this lemma we can now prove that nowhere dense sets have the following properties:

- (a) Any subset of a nowhere dense set is nowhere dense.
- (b) The union of finitely many nowhere dense sets is nowhere dense.
- (c) The closure of a nowhere dense set is nowhere dense.

Proof. (a) Suppose that A is a nowhere dense set and $X \subseteq A$. Then $X \subseteq A \implies (\overline{X})^\circ \subseteq (\overline{A})^\circ = \emptyset$. It follows immediately that X must be nowhere dense.

- (b) Consider the case of $A \cup B$ where A, B are nowhere dense sets in X . From Lemma 1.2 it suffices to show that $A^c \cap B^c$ is open and dense in X . It follows immediately that A^c and B^c are open sets. To prove density, consider some non-empty open set $U \subseteq X$, then $U_1 = U \cap A^c \neq \emptyset$. Since U_1 is open, we can say that $U_2 = U_1 \cap B^c \neq \emptyset$ is also open. Then,

$$U_2 = U \cap A^c \cap B^c \neq \emptyset$$

is open in X . Therefore, $A^c \cap B^c$ is dense, and $A \cup B$ must be nowhere dense. For a finite union of nowhere dense sets, it suffices to iteratively repeat this process to show that the union is nowhere dense.

- (c) Let A be a nowhere dense set. Then $(\overline{\overline{A}})^\circ = (\overline{A})^\circ = \emptyset$.

□

For example a point in $\mathbb{R}^n$ is nowhere dense, i.e. $\overline{\{x\}} = \{x\}$ and $\{x\}^\circ = \emptyset$. On the other hand $\mathbb{Q}$ is not a nowhere dense set because $(\overline{\mathbb{Q}})^\circ = \mathbb{R}^\circ = \mathbb{R}$. One final definition is a classification that we will use in order to make a simplification of one of the claims presented by Baire.

Definition 1.11 (G_δ set). *A set is called G_δ if it can be represented as a countable intersection of open sets.*

1.3 Baire Category Theorem

The modern presentation of Baire's Category theorem takes the following form.

Theorem 1.1 (Baire Category Theorem). *Let (X, d) be a complete metric space. If $\{O_n\}_{n=1}^\infty$ is a sequence of open and dense sets, then*

$$\bigcap_{n=1}^\infty O_n$$

is dense and uncountable in (X, d) .

Proof. Let B and $O = \bigcap_{n=1}^\infty O_n$ be an open ball in a complete metric space (X, d) . We claim that $B \cap O \neq \emptyset$. Since O_1 is dense note that $B \cap O_1 \neq \emptyset$. Since $B \cap O_1$ is open there exist an open ball $B(x_1, \epsilon_1)$ where $x_1 \in X$ and $\epsilon_1 > 0$ where

$$B(x_1, \epsilon_1) \subset B \cap O_1$$

Since O_2 is dense note again that $B(x_1, \epsilon_1) \cap O_2 \neq \emptyset$. Then there exists another open ball $B(x_2, \epsilon_2)$ where $x_2 \in X$ and $\epsilon < \frac{\epsilon_1}{2}$ where

$$B(x_2, \epsilon_2) \subset B(x_1, \epsilon_1) \cap O_1 \subset B \cap O_1 \cap O_2$$

Now suppose for some k , we have constructed

$$B(x_k, \epsilon_k) \subset B \cap O_1 \cap O_2 \cap \cdots \cap O_k$$

where $\epsilon_{k+1} < \frac{\epsilon_k}{2}$. Then $B(x_k, \epsilon_k) \cap O_{k+1} \neq \emptyset$ because O_{k+1} is dense. Since O_k is open, there exists an open ball $B(x_{k+1}, \epsilon_{k+1})$ taking $\epsilon_{k+1} < \frac{\epsilon_k}{2^{k+1}}$ where

$$B(x_{k+1}, \epsilon_{k+1}) \subset B(x_k, \epsilon_k) \cap O_{k+1} \subset B \cap O_1 \cap \cdots \cap O_{k+1}$$

Now consider the sequence of centers $\{x_n\}_{n=1}^\infty$. This sequence must be Cauchy; therefore, for all $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that for all $m, n > N$ where $m > n$ and $2^n > \frac{\epsilon_1}{\epsilon}$ and

$$x_m \in B(x_m, \epsilon_m) \subset B(x_{m-1}, \epsilon_{m-1}) \subset \cdots \subset B(x_n, \epsilon_n)$$

then, $d(x_m, x_n) < \epsilon_n < \frac{\epsilon}{2^n} < \epsilon$. Since (X, d) is a complete metric space, x_k converges to an $x^* \in X$. Note that for all $m > k$

$$x^* \in \overline{B(x_k, \epsilon_k)} \subset B(x_{k-1}, \epsilon_{k-1}) \subset B \cap O_1 \cap \cdots \cap O_{k-1}$$

Because this holds for all k as $k \rightarrow \infty$ then

$$x^* \in B \cap \bigcap_{n=1}^\infty O_n$$

Now to show that O is uncountable suppose towards contradiction that it is countable. Then

$$O = \{a_1, a_2, \dots\}$$

Define $U_n = X \setminus \{a_n\}$ which is open and dense. If $U = O \cap \bigcap_{n=1}^\infty U_n$. Then eventually $U = \emptyset$, which contradicts the assumption that U is dense. $\square$

That is if we have a dense G_δ , a countable intersection of open and dense sets, then this set is dense and uncountable. From this theorem arise the following classifications:

1st Category A set $E \subset X$ is of the first category in X if E is a countable union of nowhere dense sets. Also called a meager set.

2nd Category A set $E \subset X$ is of this category if it isn't of the first category.

Generic A set $E \subset X$ is generic if it's complement is of the first category.

The categorizations aim to develop a notion of "size" of a subset. Sets that are meager are "small" in the sense that they cannot contain an open ball. On the other hand, sets in the second category are "large". Generic sets, unlike the first and second categories, are more "frequently" occurring than sets of either category. In fact, a generic set is always a set of the second category, while a set in the second category may not be generic. For example, $\mathbb{R}_{\geq 0}$ is of the second category because it has a non-empty interior while its complement $\mathbb{R}_{< 0}$ is of the second category for the same reason. Then $\mathbb{R}$ is a second category set that isn't generic. Given that the generic sets represent the majority of sets the definition that Baire presents is fairly difficult to work with. Instead I will present the definition in terms of the more familiar G_δ classification.

Theorem 1.2. A set A is generic in X if and only if there exists a countable intersection of open dense subsets $\{U_n\}_{n=1}^\infty$ such that $\bigcap_{n=1}^\infty U_n \subseteq A$ i.e. if A admits a dense G_δ set.

Proof. Suppose that X is generic, then $X \setminus A$ is meager. This means there exists an intersection of countably many nowhere dense sets U_n such that $X \setminus A = \bigcup_{n=1}^\infty U_n$. Define $F_n := X \setminus \overline{U_n}$ it follows that F_n is open. Density follows from the identity $\overline{X \setminus A} = X$. Note that

$$\bigcap_{n=1}^\infty F_n = X \setminus \bigcap_{n=1}^\infty \overline{U_n}$$

Since $U_n \subseteq \overline{U_n}$ then,

$$\bigcap_{n=1}^\infty F_n = X \setminus \bigcap_{n=1}^\infty \overline{U_n} \subseteq X \setminus \bigcap_{n=1}^\infty U_n = A$$

Then A admits a dense G_δ subset. Now suppose that there exists a dense G_δ set such that $\bigcap_{n=1}^\infty U_n \subseteq A$ in X . Then,

$$X \setminus A \subseteq X \setminus \bigcap_{n=1}^\infty U_n = \bigcup_{n=1}^\infty X \setminus U_n$$

Now define $A_n := X \setminus U_n$. Note that

$$(\overline{A_n})^\circ = (\overline{X \setminus U_n})^\circ = (X \setminus X^\circ)^\circ = \emptyset$$

then A_n is nowhere dense. It follows that

$$\bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} X \setminus U_n$$

By assumption, we can pick U_n such that

$$X \setminus A = X \setminus \bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} A_n$$

□

With this definition, we can now prove that every open dense set is generic.

Lemma 1.3. *Any open dense set in a space X is generic.*

Proof. Let X be an open dense set then there exists a countable sequence of dense sets U_n such that each $U_n = X$ then, it immediately follows that $\bigcap_{n=1}^{\infty} U_n \subseteq X$. □

The key result presented by Baire is that the real line is of the second category. The primary ingredient in his argument is the fact that the real line is complete. The reliance on completeness allows us to generalize these categories to any metric space.

Theorem 1.3. *A complete metric space X is of the second category, X cannot be written as the countable union of nowhere dense sets.*

Proof. Assume towards contradiction that $X = \bigcup_{n=1}^{\infty} F_n$ where each F_n is nowhere dense. Replace each F_n with its closure and view X as closed. Since F_1 is closed there exists an open ball B_{r_1} or radius r_1 such that $\overline{B_{r_1}} \subset F_1^c$. Since F_2 is nowhere dense then B_1 cannot be entirely contained in F_2 otherwise the interior would be non-empty. Since F_2 is closed then there exists a ball B_{r_2} such that $r_2 < \frac{r_1}{2}$ and $\overline{B_{r_2}} \subset B_1$ in F_2^c . Iteratively repeating this process furnishes us with a sequence of open balls with the following properties.

- (i) Each radius $r_n < \frac{r_{n-1}}{2} < \dots < \frac{r_1}{2}$ then $r_n < \frac{r_1}{2^m}$. Then as $n \rightarrow \infty$ the radius $r_n \rightarrow 0$.
- (ii) Given that $\overline{B_{r_{n+1}}} \subset B_{r_n}$ it follows that $B_{r_{n+1}} \subset B_{r_n}$.
- (iii) Since $\overline{B_{r_n}} \subset F_n^c$ then $F_n \cap B_{r_n} = \emptyset$.

Now pick a point $x_n \in B_{r_n}$ for each n . The sequence $\{x_n\}_{n=1}^{\infty}$ is Cauchy following from (i) and (ii). From the completeness of X , the sequence $x_n \rightarrow x$ implying that $x \in \overline{B_{r_n}}$ for all n . But (iii) implies that $x \notin F_n$ for all n . This contradicts our assumption therefore X is of the second category. □

Immediately following this result we get a lemma allowing us to classify dense sets in a complete metric space.

Lemma 1.4. *In a complete metric space, a generic set is dense.*

Proof. Let X be a complete metric space. Suppose towards contradiction that $E \subset X$ is generic but not dense. It follows that there exists a closed ball $\overline{B} \subset E^c$. Since $E = \bigcup_{n=1}^{\infty} F_n$, where each F_n is nowhere dense. Then $\overline{B} = \bigcup_{n=1}^{\infty} (F_n \cap \overline{B})$. Consider $\overline{B}$ as a restriction of the metric space X . Since $\overline{B}$ inherits the completeness of X and $F_n \cap \overline{B}$ is not nowhere dense then we can contradict the previous theorem. □

1.3.1 Simple Examples

A quick and interesting example is to think about the Cantor Ternary Set C . As a reminder this set is constructed iteratively by removing the open middle third of all intervals. We can make this more precise by considering the recursive definition

$$C_0 := [0, 1] \quad \text{and} \quad C_n := \left(\frac{C_{n-1}}{3}\right) \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3}\right)$$

where the set itself is defined $C := \lim_{n \rightarrow \infty} C_n = \bigcap_{n=1}^{\infty} C_n$. It is a well known fact that the Cantor set is closed and its interior is empty. Then, in a complete metric space $\mathbb{R}$ the Cantor set is meager. On the other hand, the Cantor Set, in itself, is not meager, primarily because it doesn't form a complete metric space.

2 Application of Baire Category Theorem

2.1 Existence of Continuous but Nowhere Differentiable Functions

Up until the early nineteenth century, many mathematicians believed that a continuous function must be differentiable. Now most undergraduate mathematics students know this is, unfortunately, not the case. Weierstrass was the first to contradict the claim, introducing a particular Fourier Series

$$f(x) = \sum_{n=0}^{\infty} a^n \cos(b^n \pi x)$$

where $0 < a < 1$, b is a positive odd integer, and $ab > 1 + \frac{3}{2}\pi$. The presentation of the Baire Category theorem, introduced nearly a century later, provides a new avenue to prove that there exists a continuous nowhere differentiable function. We will plant ourselves in the metric space of the set of continuous functions $C[0, 1]$ with the norm $\|f\| = \sup_{x \in [0, 1]} |f(x)|$ inducing the metric $d(f, g) = \|f - g\|$. In this metric space we will work towards proving the following theorem.

Theorem 2.1. *The set of functions $C[0, 1]$ that are nowhere differentiable is generic.*

Conversely, the set of continuous, everywhere differentiable functions is meager. We can interpret this claim as

The set containing continuous everywhere differentiable functions on $[0, 1]$ is "smaller" than the set of all continuous functions on $[0, 1]$.

To tackle this theorem we will show that the set of functions that are continuous, and differentiable on $[0, 1]$ at least at one point, denoted D . Define the following set

$$E_N := \{f \in C[0, 1] : \exists \hat{x} \in (0, 1) \text{ s.t. } |f(x) - f(\hat{x})| \leq N|x - \hat{x}| \forall x \in [0, 1]\}$$

To prove our theorem we need to prove the following statements:

- (i) $D \subset \bigcup_{n=1}^{\infty} E_n$.
- (ii) E_n is closed.
- (ii) E_n has an empty interior.

2.1.1 Proof of (i)

Proof. Let $f \in C[0, 1]$ be differentiable at least at one point. Then there exists a point $\hat{x} \in [0, 1]$ such that the limit

$$f'(\hat{x}) = \lim_{h \rightarrow 0} \left| \frac{f(\hat{x} + h) - f(\hat{x})}{h} \right|$$

exists. Then there exists a $\delta > 0$ and an $M > 0$ such that when $|x - \hat{x}| < \delta$ then $|f(x) - f(\hat{x})| \leq M|x - \hat{x}|$. Since our definition of n requires a global bound define

$$N = \max \left\{ M, \frac{2\|f\|}{\delta} \right\} < \infty$$

With n we can extend our bound

$$|f(x) - f(\hat{x})| \leq N|x - \hat{x}| \quad \forall x \in [0, 1]$$

Therefore $f \in E_N$. Then if f is differentiable it follows that $f \in E_N$ for some N i.e. $D \subset \bigcup_{n=1}^{\infty} E_n$. $\square$

2.1.2 Proof of (ii)

Proof. Suppose that $f_n(x)$ is a Cauchy Sequence of functions in $C[0, 1]$ which converge to some $f(x) \in C[0, 1]$. Let $\hat{x}_n$ be a sequence of points in $[0, 1]$ such that $f(\hat{x}_n)$ is well defined for all n . Pick a subsequence $\hat{x}_{n_k}$ that converges to a point $\hat{x} \in [0, 1]$. Then,

$$|f(x) - f(\hat{x})| = |f(x) - f_{n_k}(x) + f_{n_k}(x) - f_{n_k}(\hat{x}) + f_{n_k}(\hat{x}) - f(\hat{x})| \quad (8)$$

$$\leq |f(x) - f_{n_k}(x)| + |f_{n_k}(x) - f_{n_k}(\hat{x})| + |f_{n_k}(\hat{x}) - f(\hat{x})| \quad (9)$$

Since $f_{n_k}(x) \rightarrow f(x)$ then there exists a maximal K such that for all $k \geq K$, $|f(x) - f_{n_k}(x)| < \frac{\epsilon}{2}$. Simplifying our relation to

$$|f(x) - f_{n_k}(x)| + |f_{n_k}(x) - f_{n_k}(\hat{x})| + |f_{n_k}(\hat{x}) - f(\hat{x})| < \epsilon + |f_{n_k}(x) - f_{n_k}(\hat{x})|$$

Utilizing the subsequence $\hat{x}_{n_k}$ and the fact that $f_{n_k} \in E_n$ then,

$$|f_{n_k}(x) - f_{n_k}(\hat{x})| = |f_{n_k}(x) - f_{n_k}(\hat{x}_{n_k}) + f_{n_k}(\hat{x}_{n_k}) - f_{n_k}(\hat{x})| \quad (10)$$

$$\leq |f_{n_k}(x) - f_{n_k}(\hat{x}_{n_k})| + |f_{n_k}(\hat{x}_{n_k}) - f_{n_k}(\hat{x})| \quad (11)$$

$$< N|x - \hat{x}_{n_k}| + N|\hat{x}_{n_k} - \hat{x}| \quad (12)$$

Finally, we see that

$$|f(x) - f(\hat{x})| \leq \epsilon + N|x - \hat{x}_{n_k}| + N|\hat{x}_{n_k} - \hat{x}| < N|x - \hat{x}|$$

therefore $f(x) \in E_n$ and each E_n is closed. $\square$

2.1.3 Proof of (iii)

This part represents the most difficult portion of the proof. To show that the interior is empty, we will define a particular subspace P of $C[0, 1]$ consisting of all piecewise linear functions in $C[0, 1]$. We will further decompose this subspace into two portions. Let P_M denote the set of piecewise linear functions whose slope is $\geq M$ and $\leq -M$. In order to show that E_N has an empty interior we will show that P_M is dense in $C[0, 1]$. To do this we first need to show that any continuous function can be approximated with arbitrary precision by a piecewise linear function.

Theorem 2.2. *If $f \in C[a, b]$ is a continuous function on the interval $[a, b]$, then for all $\epsilon > 0$ there exists a piecewise-linear function p such that*

$$\sup_{x \in [a, b]} |f(x) - p(x)| < \epsilon$$

Proof. Since $[a, b]$ is compact, f is uniformly continuous on $[a, b]$. Then for all $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x, y \in [a, b]$ with $|x - y| < \delta$, we have $|f(x) - f(y)| < \epsilon$. Fix an $N \geq 1$ where $\frac{(b-a)}{N} < \delta$ and subdivide $[a, b]$ into N evenly spaced sub-intervals $[a, x_1], [x_2, x_3], \dots, [x_{N-1}, b]$. Let g be a piecewise linear function that agrees with f on points $a, x_1, \dots, x_{N-1}, b$. By construction, each interval has a width less than δ . Now let x be any point in $[a, b]$. This point must be in some sub-interval $[a', b']$ of the list of constructed sub-intervals above. Because the interval has width less than δ , we know that $|f(x) - f(a')| < \epsilon$ and $|f(x) - f(b')| < \epsilon$. Since g is linear on $[a', b']$, there exists some $t \in [0, 1]$ such that $g(x) = tf(a') + (1 - t)f(b')$. From the triangle inequality, it follows

$$|f(x) - g(x)| \leq t|f(x) - f(a')| + (1 - t)|f(x) - f(b')| < \epsilon$$

$\square$

With this result, we are now ready to prove the following theorem.

Theorem 2.3. *The set of piecewise linear functions of slope $+M$ and $-M$ denoted P_M is dense in $C[0, 1]$.*

Proof. For any continuous function $f \in C[0, 1]$, theorem (2.2) guarantees that there exists a piecewise linear function g on n even subdivisions of $[0, 1]$ such that $\|f - g\| < \epsilon$. Now we will show that g can be arbitrarily approximated with piecewise functions in P_M . Since g is linear, then $g(x) = ax + b$ for all $x \in [0, \frac{1}{n}]$, now we can define two functions

$$\varphi_\epsilon = g(x) + \epsilon \quad \text{and} \quad \psi_\epsilon = g(x) - \epsilon$$

Then we can choose a line segment of slope M in P_M such that we are traveling from φ_ϵ to ψ_ϵ and another of slope $-M$ from ψ_ϵ to φ_ϵ . This gives us a piecewise function $h(x)$ such that $\varphi_\epsilon(x) \leq h(x) \leq \psi_\epsilon(x)$ for all $x \in [0, \frac{1}{n}]$. It follows from construction that $|h(x) - g(x)| < \epsilon$ in $[0, \frac{1}{n}]$. Now we can repeat this argument for each interval $[\frac{k}{n}, \frac{k+1}{n}]$ in order to get a function $h(x) \in P_M$. In particular, $\|g - h\| < \epsilon$ then

$$\|f - h\| \leq \|f - g\| + \|g - h\| = 2\epsilon$$

Then clearly P_M is dense in $C[0, 1]$. □

With this result we can now show that E_N has no interior points. Assume towards contradiction that E_N has interior points. Then there must exist an open ball $B(f, \epsilon)$, where $f \in C[0, 1]$ entirely contained in E_N . Fix $M > N$, from results above there exists a function $h \in P_M$ such that $\|f - h\| < \epsilon$. Then $h \in B(f, \epsilon)$ and $h \notin E_N$ implying that $B(f, \epsilon)$ is not entirely contained in E_N . Therefore E_N has no interior points.

3 References

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